Experiment 10: Closest Pair of Points – Brute Force

Aim : To find the pair of points having the minimum Euclidean distance using the brute-force approach.

Algorithm

1. Read the number of points and their coordinates.
2. Initialize the minimum distance to infinity.
3. Compare every pair of distinct points.
4. Calculate the Euclidean distance between each pair.
5. Update the minimum distance and closest pair whenever a smaller distance is found.
6. Display the closest pair and minimum distance.

Python Program

```
import math

n = int(input("Enter number of points: "))

points = []

for i in range(n):

    x, y = map(int, input(f"Enter coordinates of point {i + 1}: ").split())

    points.append((x, y))

if n < 2:

    print("At least two points are required.")

else:

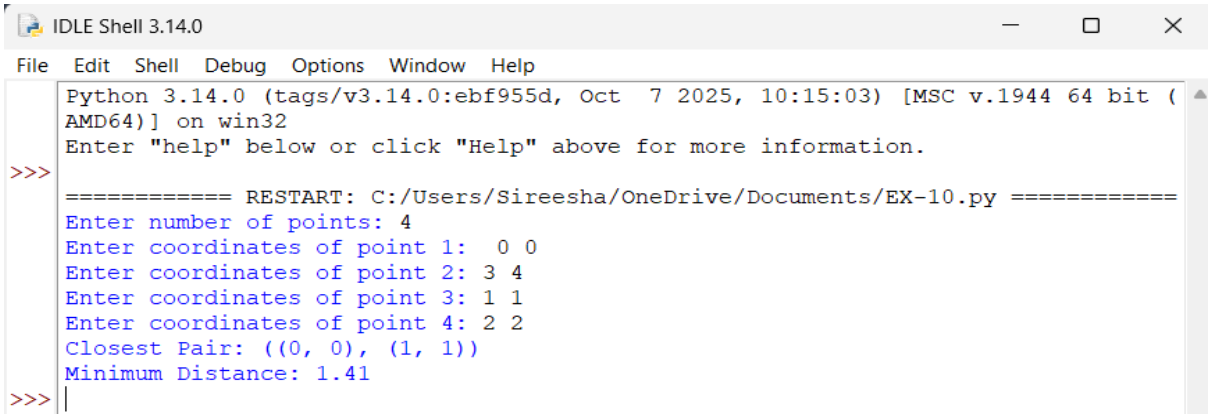
    min_distance = float("inf")
```

```
    closest_pair = None
    for i in range(n):
        for j in range(i + 1, n):
            x1, y1 = points[i]
            x2, y2 = points[j]
            distance = math.sqrt(
                (x2 - x1) ** 2 + (y2 - y1) ** 2
            )
            if distance < min_distance:
                min_distance = distance
                closest_pair = (points[i], points[j])
    print("Closest Pair:", closest_pair)
    print("Minimum Distance:", round(min_distance, 2))
```

Input

```
Enter number of points: 4
Enter coordinates of point 1: 0 0
Enter coordinates of point 2: 3 4
Enter coordinates of point 3: 1 1
Enter coordinates of point 4: 2 2
```

Output



Result

Thus, the closest pair of points and their minimum Euclidean distance were successfully found using the brute-force approach.